

4. Unit step signal and unit impulse signal

Main Content

Singularity signals

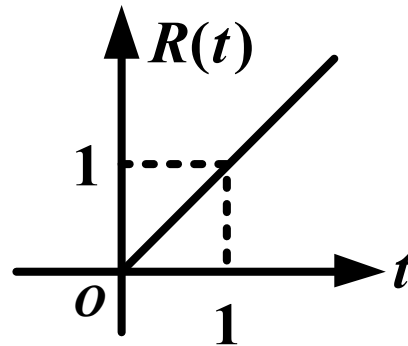
1. Unit ramp signal
2. Unit step signal
3. Unit impulse signal
4. Doublet signal

4. Unit step signal and unit impulse signal

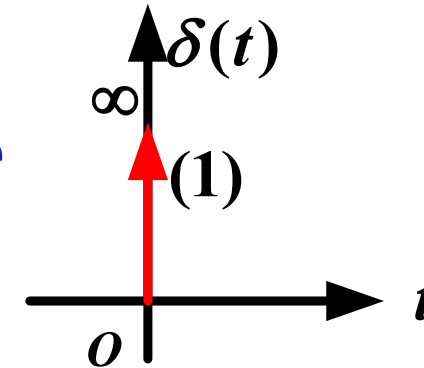
1. Singularity signals

Functions that themselves have discontinuity points (jump discontinuities) or whose derivatives/integrals contain discontinuities are collectively referred to as singular signals or singularity functions

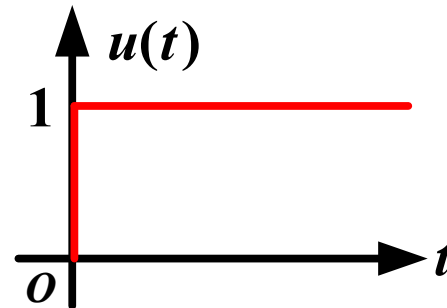
Unit ramp
signal



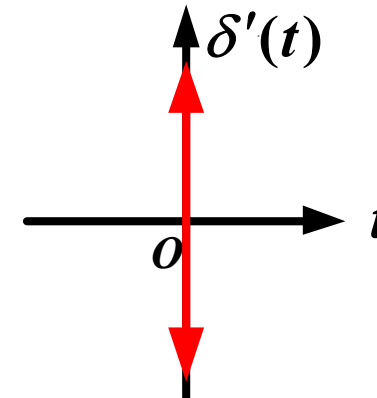
Unit impulse
signal



Unit step
signal



Doublet
signal

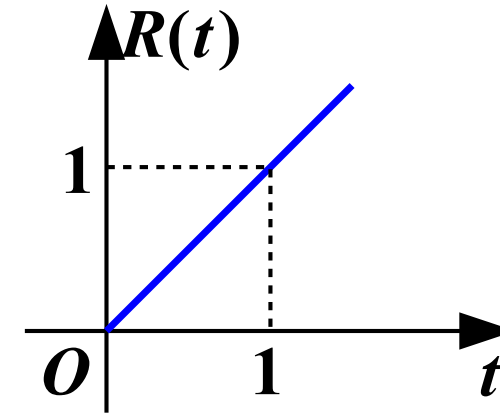


4. Unit step signal and unit impulse signal

2. Unit ramp signal

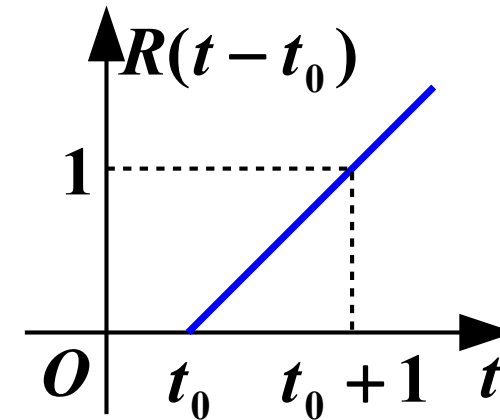
(1) Definition

$$R(t) = \begin{cases} 0 & t < 0 \\ t & t \geq 0 \end{cases}$$



(2) Delayed unit ramp signal

$$R(t - t_0) = \begin{cases} 0 & t < t_0 \\ t - t_0 & t \geq t_0 \end{cases}$$



From the argument $t - t_0 = 0$, the starting point is determined to be t_0

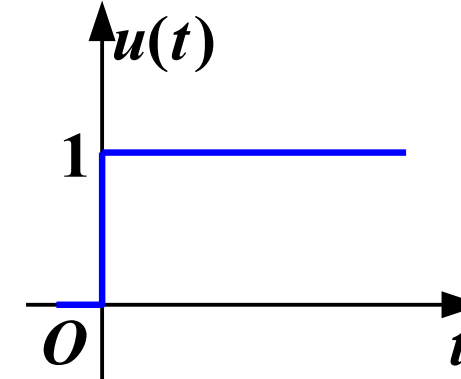
4. Unit step signal and unit impulse signal

3. Unit step signal

(1) Definition

$$u(t) = \begin{cases} 0 & t < 0 \\ 1 & t > 0 \end{cases}$$

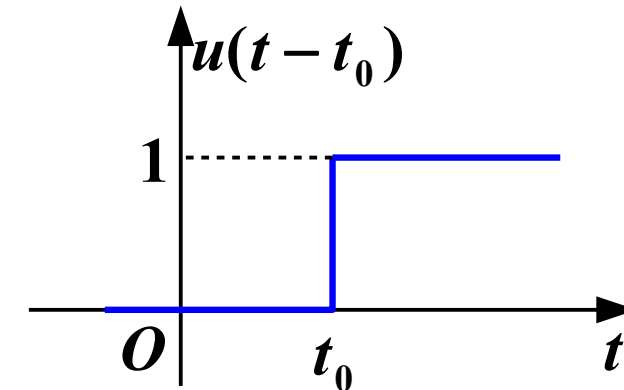
At $t=0$, the function is either undefined or equals $1/2$



When the argument > 0 , the function value is 1, When the argument < 0 , the function value is 0

(2) Delayed unit step signal

$$u(t - t_0) = \begin{cases} 0 & t < t_0 \\ 1 & t > t_0 \end{cases}, t_0 > 0$$



When the argument $(t-t_0)=0$ (i.e., at $t=t_0$), the function has a discontinuity (jump point)

4. Unit step signal and unit impulse signal

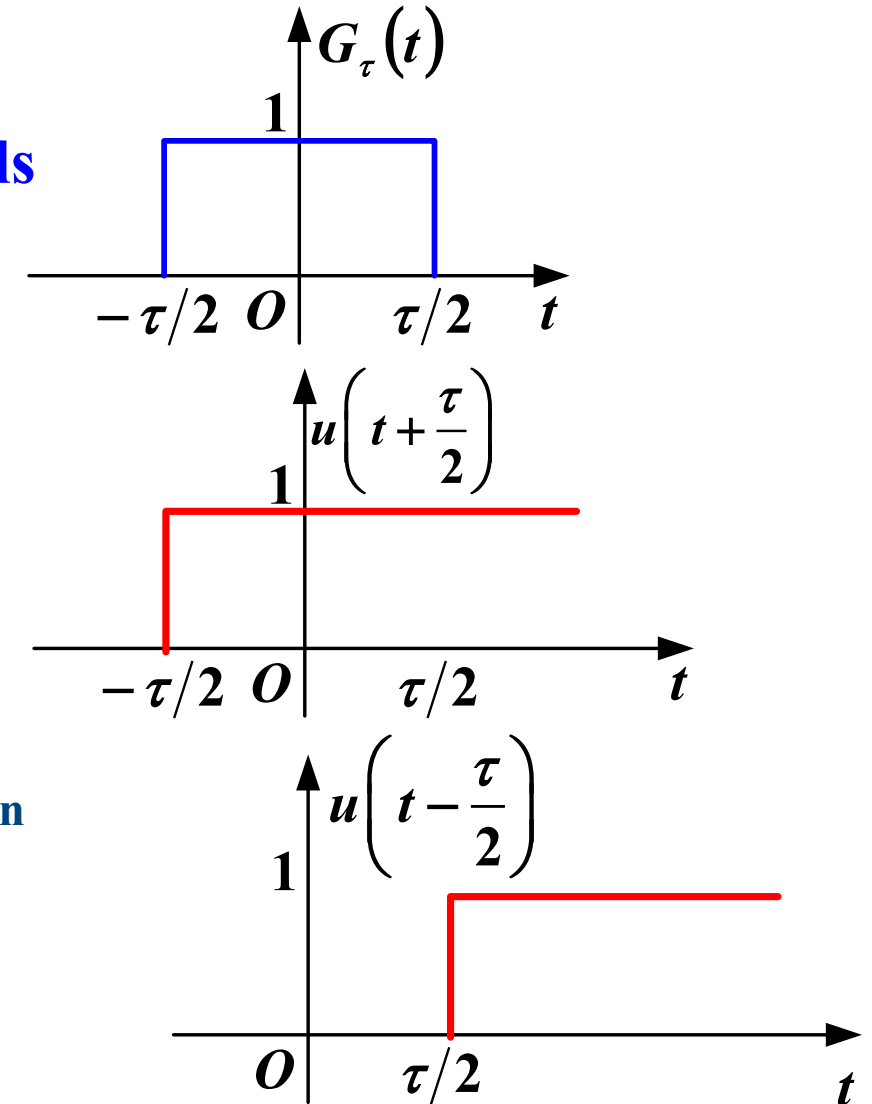
3. Unit step signal

(3) Using unit step functions to represent other signals

Gate Function (Rectangular Window Function)

$$G_{\tau}(t) = u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right)$$

When any signal is multiplied by a gate function, only the portion within the "gate" remains



4. Unit step signal and unit impulse signal

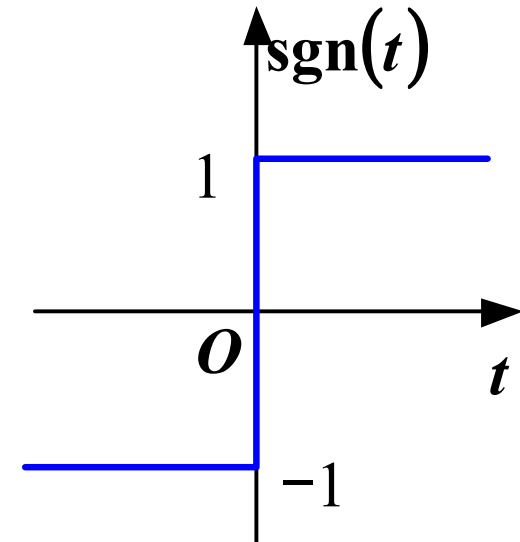
3. Unit step signal

(3) Using unit step functions to represent other signals

Signum Function

$$\text{sgn}(t) = \begin{cases} 1, & t > 0 \\ -1, & t < 0 \end{cases}$$

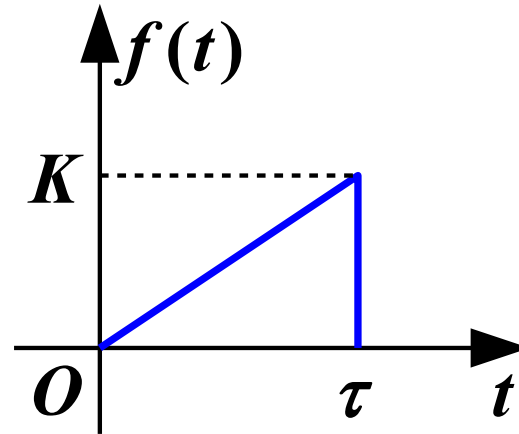
$$\begin{aligned} \text{sgn}(t) &= -u(-t) + u(t) \\ &= 2u(t) - 1 \end{aligned}$$



4. Unit step signal and unit impulse signal

3. Unit step signal

(3) Using unit step functions to represent other signals



$$f(t) = \frac{K}{\tau} t [u(t) - u(t - \tau)]$$

4. Unit step signal and unit impulse signal

4. Unit impulse signal

Concept Introduction

Definition 1: Limit of Pulse Signals

Definition 2: Dirac Function

Definition 3: Generalized Function Definition

4. Unit step signal and unit impulse signal

(1) Concept Introduction

- **Impulsive Force in Mechanics**, A force that arises during collisions between objects, characterized by a sudden increase followed by rapid dissipation
- **Key Features**, Acts over an extremely short duration, can attain very large magnitudes
- **Examples**, Sports mechanics, striking a ping-pong ball, kicking a soccer ball
- **Analogous Concepts in Other Fields**, Particle dynamics, Point charge impulse, Point light source intensity spikes, Impulse signals



4. Unit step signal and unit impulse signal

(2) Definition 1: Limit of Pulse Signals

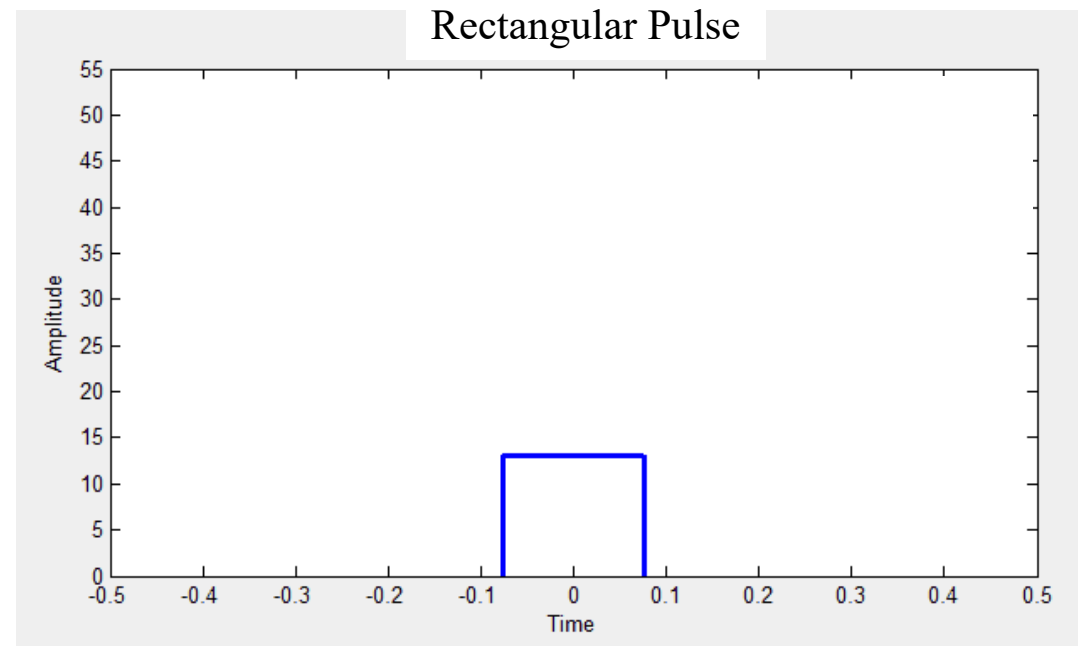
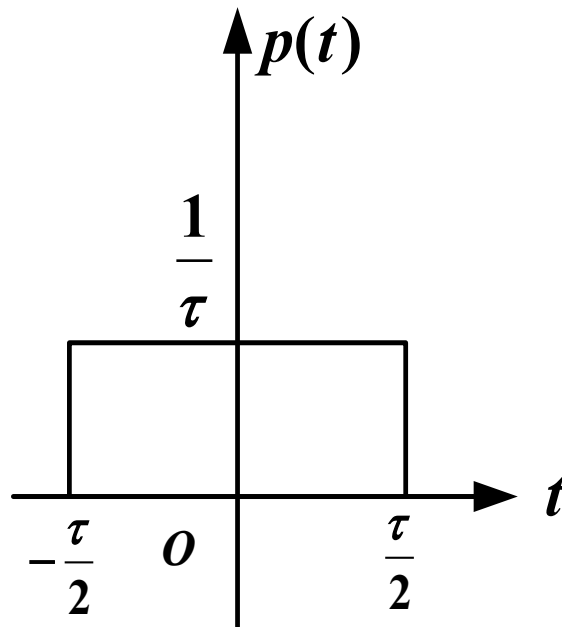
$$p(t) = \frac{1}{\tau} \left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$$

$\tau \rightarrow 0$

Area 1

Pulse Width↓

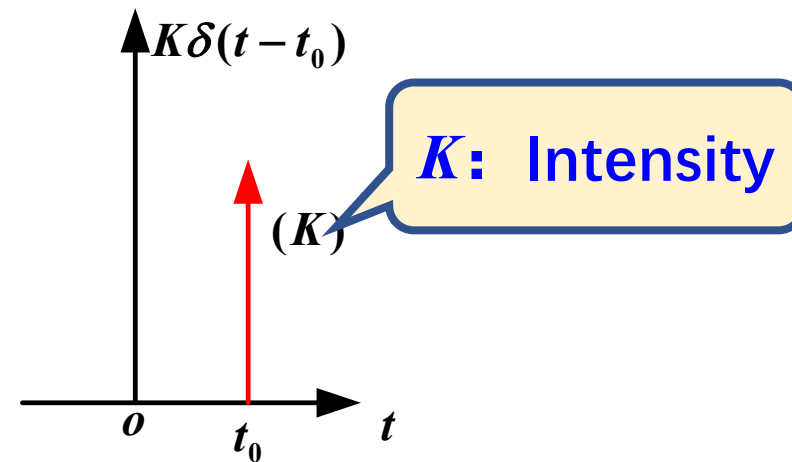
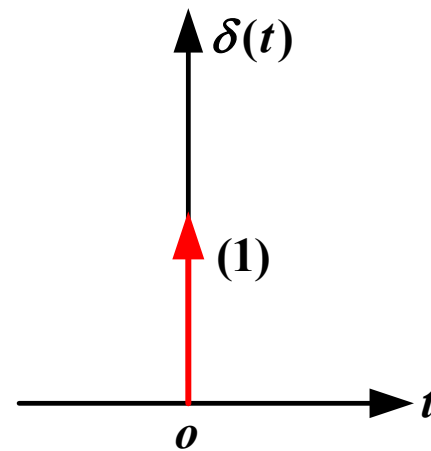
Pulse Height↑



4. Unit step signal and unit impulse signal

(2) Definition 1: Limit of Pulse Signals

$$\delta(t) = \lim_{\tau \rightarrow 0} p(t) = \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left[u\left(t + \frac{\tau}{2}\right) - u\left(t - \frac{\tau}{2}\right) \right]$$



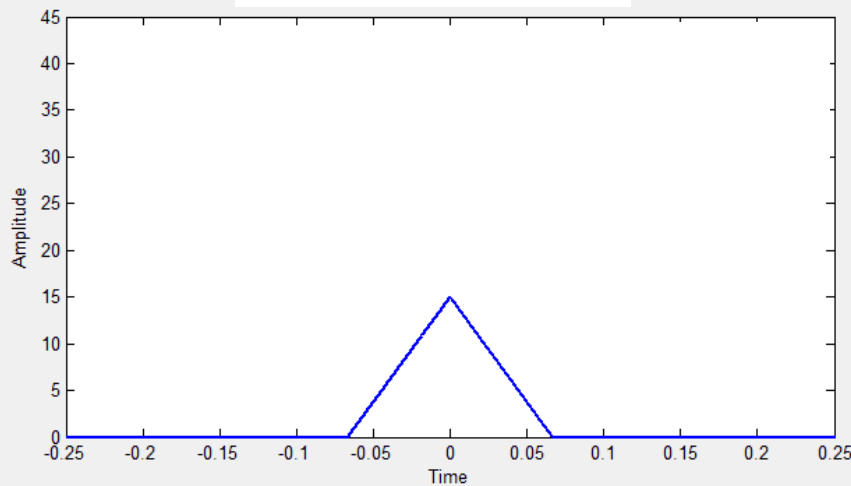
Triangular pulse, double-sided exponential pulse, Gaussian pulse, and the sinc function taking the limit as $t \rightarrow 0$ can all be considered as impulse functions

4. Unit step signal and unit impulse signal

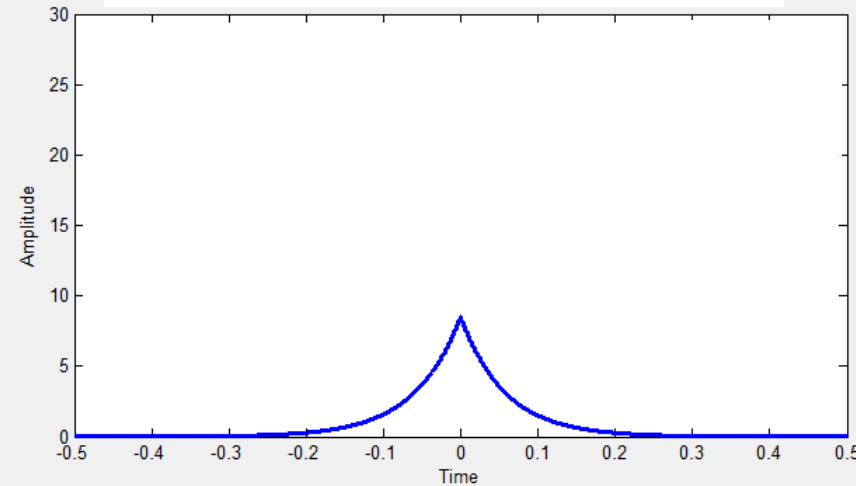
(2) Definition 1: Limit of Pulse Signals

$$\delta(t) = \lim_{\tau \rightarrow 0} \frac{1}{\tau} \left(1 - \frac{|t|}{\tau} \right) \quad (|t| \leq \tau) \quad \delta(t) = \lim_{\tau \rightarrow 0} \left(\frac{1}{2\tau} e^{-\frac{|t|}{\tau}} \right)$$

Triangular pulse



double-sided exponential pulse



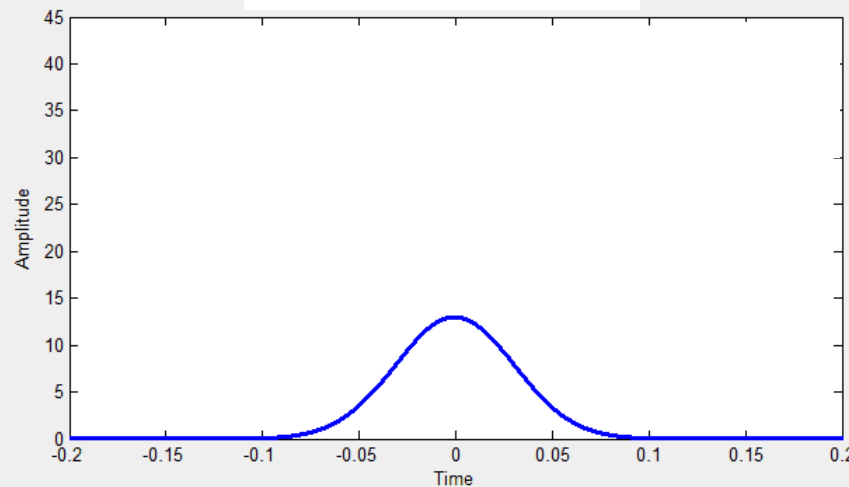
4. Unit step signal and unit impulse signal

(2) Definition 1: Limit of Pulse Signals

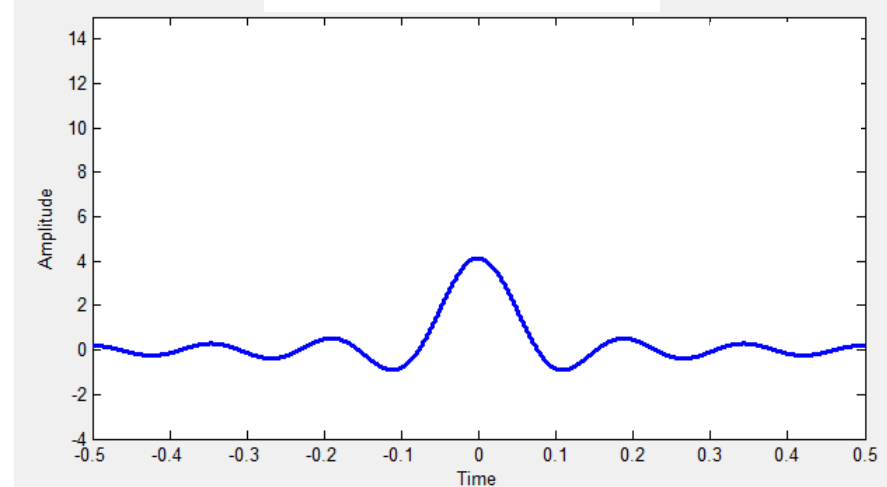
$$\delta(t) = \lim_{\tau \rightarrow 0} \left[\frac{1}{\tau} e^{-\pi \left(\frac{t}{\tau} \right)^2} \right]$$

$$\delta(t) = \lim_{k \rightarrow \infty} \left[\frac{k}{\pi} \text{Sa}(kt) \right]$$

Gaussian pulse



Sinc function

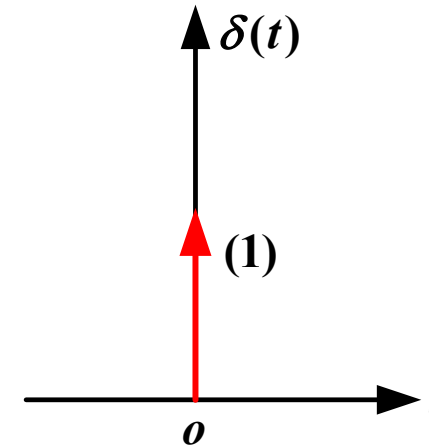


4. Unit step signal and unit impulse signal

(3) Definition 2: Dirac Function

$$\begin{cases} \int_{-\infty}^{\infty} \delta(t) dt = 1 \\ \delta(t) = 0, (t \neq 0) \end{cases}$$

$$\int_{-\infty}^{\infty} \delta(t) dt = \int_{0_-}^{0_+} \delta(t) dt = 1$$



- The function is non-zero **only** at $t = 0$
- Its integral equals **1**
- $t = 0$, $\delta(t) \rightarrow \infty$, the function is **unbounded**

(3) Definition 2: Dirac Function



Paul Adrien Maurice Dirac

August 8, 1902 – October 20, 1984

British theoretical physicist, one of the founders of quantum mechanics

Nobel Prize in Physics (1933) for the Dirac equation, which theoretically predicted the existence of the positron (antimatter)

Key Contributions

Quantum field theory: Pioneered foundational work in quantum electrodynamics (QED)

Gravitation: Advanced theories in gravity and quantization of gravitational fields.

Mathematical physics: Introduced bra-ket notation, delta function formalism, and magnetic monopole concepts.

Legacy: Regarded as one of the most influential physicists of the 20th century, bridging quantum theory and relativity

(3) Definition 3: Generalized Function Definition

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)$$

Test Functions

Distributions (Generalized Functions)

Requirements for Test Functions, $f(t)$:

- Must be continuous
- Must possess continuous derivatives of all orders
- And $f(t) \neq 0$, $a < t < b$, $(-\infty < a, b < \infty)$

$\delta(t)$ is entirely characterized by its action in integrals

4. Unit impulse signal-Properties

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)$$

To meet the requirements of signal analysis, the unit impulse function (Dirac delta function) was constructed as a generalized function (distribution). Temporally, it can be treated as a continuous-time signal in operations, as it adheres to certain rules of continuous-time signal processing. However, being a distribution, it possesses unique properties

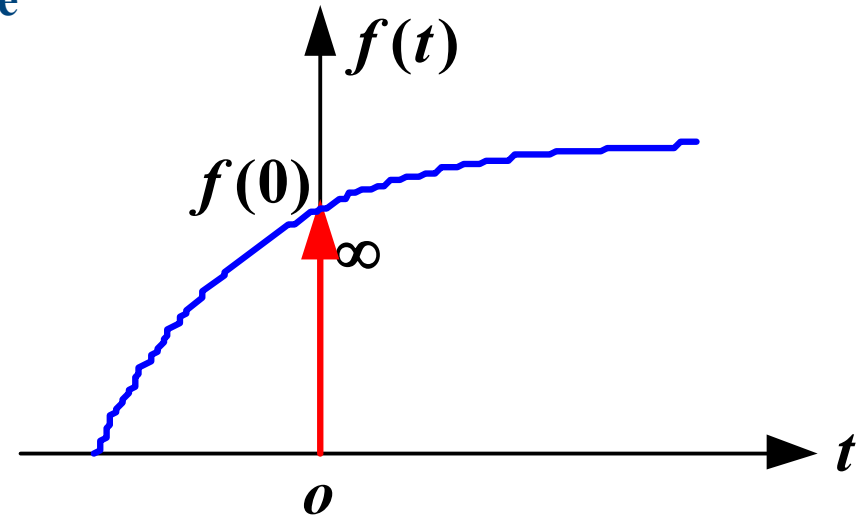
4. Unit step signal and unit impulse signal

(1) Sifting Property

If $f(t)$ is continuous at $t=0$ and bounded everywhere, then the unit impulse signal (Dirac delta function) can be rigorously defined through its action on $f(t)$

$$\delta(t)f(t) = f(0)\delta(t)$$

$$\int_{-\infty}^{\infty} \delta(t)f(t)dt = f(0)$$



This definition aligns with the theory of distributions (generalized functions), where $\delta(t)$ is characterized by its integral property rather than pointwise values

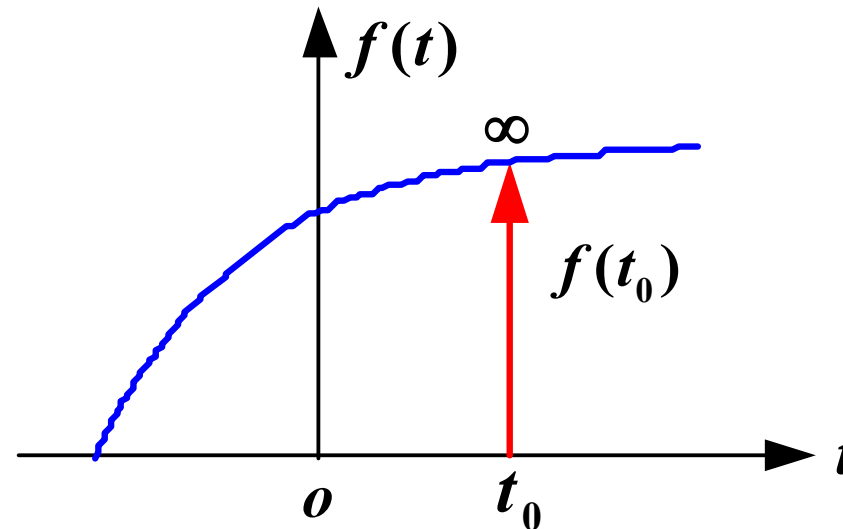
4. Unit step signal and unit impulse signal

(1) Sifting Property

Shifted

$$\delta(t - t_0) f(t) = f(t_0) \delta(t - t_0)$$

$$\int_{-\infty}^{\infty} \delta(t - t_0) f(t) dt = f(t_0)$$



4. Unit step signal and unit impulse signal

(2) Parity

- According to Definition 1, the rectangular pulse itself is an even function, so its limit is also an even function
- The parity property is demonstrated through the sampling property
- When proving the parity property, the focus is on the action of this function, i.e., the result of **its interaction with other function**

$$\int_{-\infty}^{\infty} \delta(t) f(t) dt = f(0)$$

$$\begin{aligned} \int_{-\infty}^{\infty} \delta(-t) f(t) dt & \stackrel{-t=\tau}{=} \int_{\infty}^{-\infty} \delta(\tau) f(-\tau) d(-\tau) \\ & = \int_{-\infty}^{\infty} \delta(\tau) f(-\tau) d\tau = f(0) \end{aligned}$$

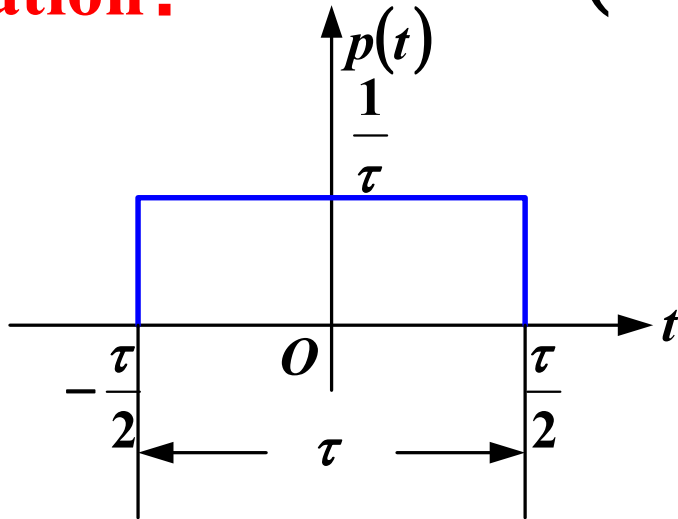
Thus $\delta(t) = \delta(-t)$

4. Unit step signal and unit impulse signal

(3) Time Scaling

Explanation:

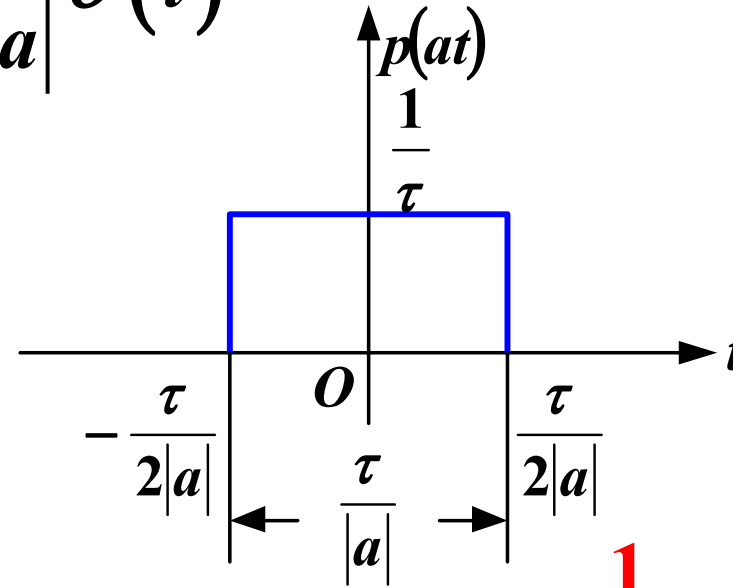
$$\delta(at) = \frac{1}{|a|} \delta(t)$$



$p(t)$ area of 1,

$\tau \rightarrow 0,$

$p(t) \rightarrow \delta(t),$



$p(at)$ area of $\frac{1}{|a|},$

$\tau \rightarrow 0,$

$p(at) \rightarrow \frac{1}{|a|} \delta(t)$

4. Unit step signal and unit impulse signal

(3) Time Scaling-Prove

Analysis: The proof is based on demonstrating the equality of the integral values of both sides multiplied by $f(t)$, considering two cases: $a > 0$ and $a < 0$

$$\int_{-\infty}^{\infty} \delta(at) f(t) dt \stackrel{?}{=} \int_{-\infty}^{\infty} \frac{1}{|a|} \delta(t) f(t) dt$$

(1) $a > 0$, let $at = \tau$, $t = \frac{\tau}{a}$

$$\int_{-\infty}^{\infty} \delta(at) f(t) dt = \int_{-\infty}^{\infty} \delta(\tau) f\left(\frac{\tau}{a}\right) d\left(\frac{\tau}{a}\right) = \frac{1}{a} f(0)$$

While $\int_{-\infty}^{\infty} \frac{1}{|a|} \delta(t) f(t) dt = \frac{1}{a} f(0)$

$\therefore$ Both sides are equal

4. Unit step signal and unit impulse signal

(3) Time Scaling-Prove

$$a < 0, \text{ let } -|a|t = \tau, t = -\frac{\tau}{|a|} \begin{cases} t : & -\infty \rightarrow +\infty \\ \tau : & +\infty \rightarrow -\infty \end{cases}$$

$$\int_{-\infty}^{+\infty} \delta(-|a|t) f(t) dt = \int_{+\infty}^{-\infty} \delta(\tau) f\left(-\frac{1}{|a|}\tau\right) d\left(-\frac{1}{|a|}\tau\right)$$

$$= \frac{1}{|a|} \int_{-\infty}^{+\infty} \delta(\tau) f\left(-\frac{1}{|a|}\tau\right) d\tau$$

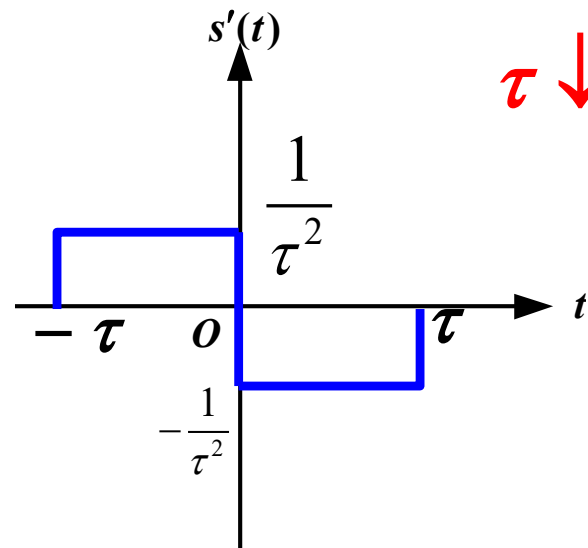
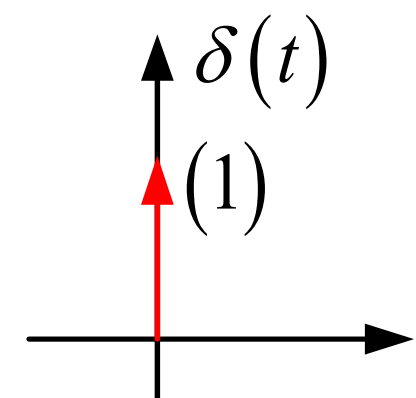
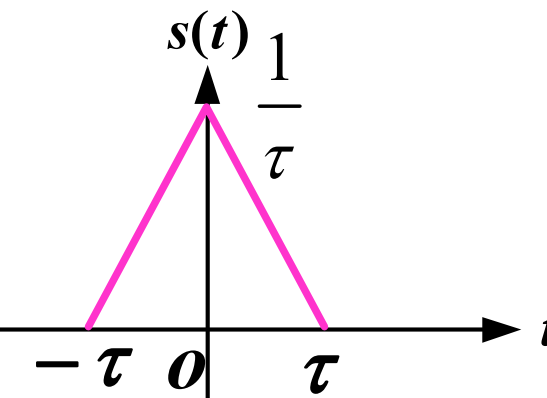
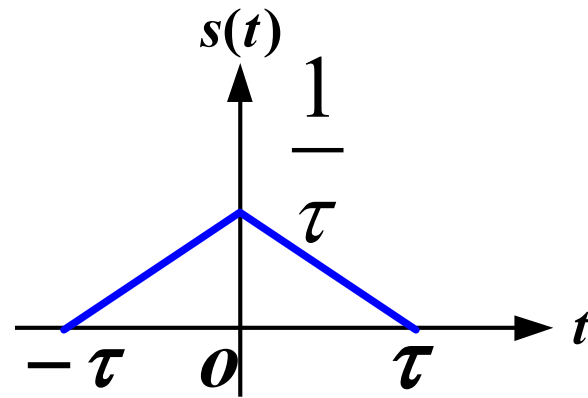
$$= \frac{1}{|a|} f(0)$$

$$\text{While } \int_{-\infty}^{+\infty} \frac{1}{|a|} \delta(t) f(t) dt = \frac{1}{|a|} f(0)$$

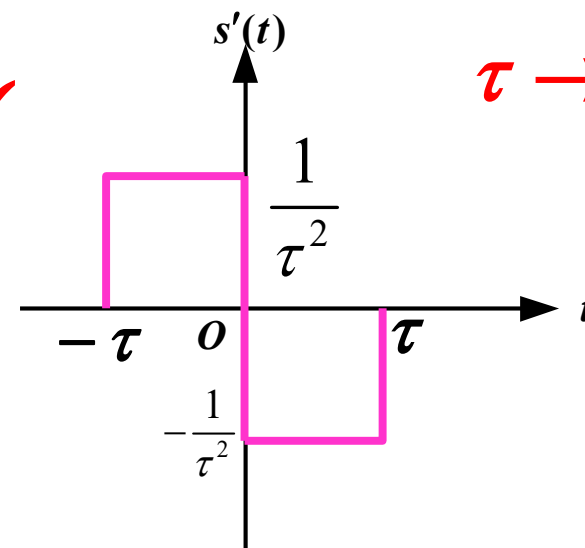
4. Unit step signal and unit impulse signal

5. Doublet signal

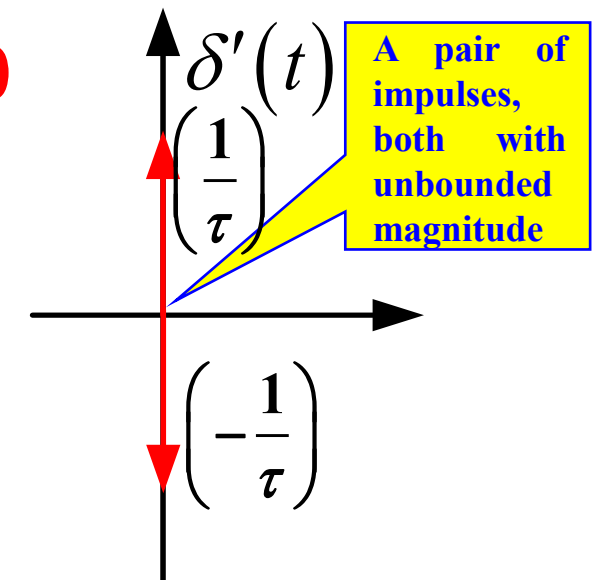
$$\delta'(t) = \frac{d\delta(t)}{dt}$$



$\tau \downarrow$



$\tau \rightarrow 0$



4. Unit step signal and unit impulse signal

5. Doublet signal

$$\textcircled{1} \int_{-\infty}^{\infty} \delta'(t) f(t) dt = -f'(0)$$

Proof: Using integration by parts

$$\begin{aligned} \int_{-\infty}^{\infty} \delta'(t) f(t) dt &= \int_{-\infty}^{\infty} f(t) d\delta(t) \\ &= f(t)\delta(t) \Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} \delta(t) df(t) \\ &= - \int_{-\infty}^{\infty} f'(t) \delta(t) dt = -f'(0) \end{aligned}$$

Shift

$$\int_{-\infty}^{\infty} \delta'(t - t_0) f(t) dt = -f'(t_0)$$

Generalization

$$\int_{-\infty}^{\infty} \delta^n(t) f(t) dt = (-1)^n f^n(0)$$

4. Unit step signal and unit impulse signal

5. Doublet signal

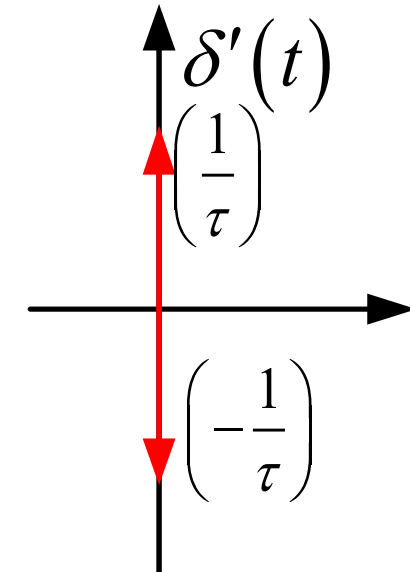
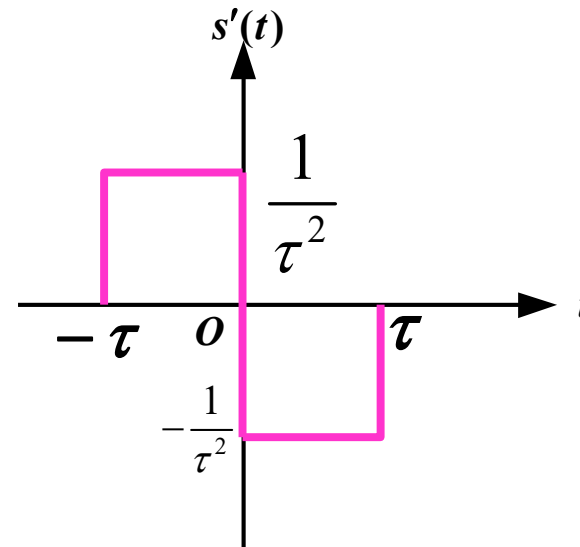
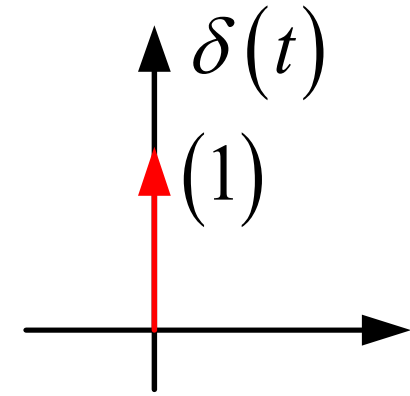
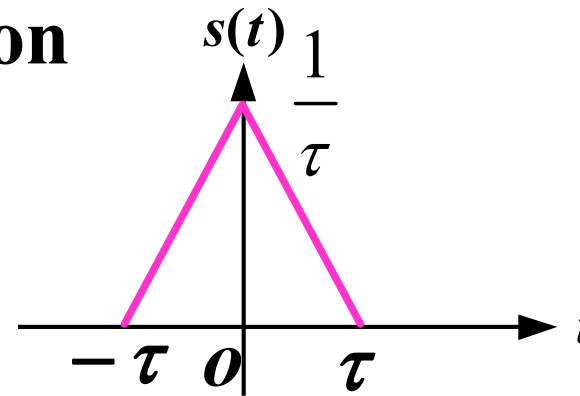
② $\delta'(t)$ is an odd function

$$\delta'(-t) = -\delta'(t)$$

$$\delta'(t_0 - t) = -\delta'(t - t_0)$$

$$\int_{-\infty}^{\infty} \delta'(\tau) d\tau = 0$$

$$\int_{-\infty}^t \delta'(\tau) d\tau = \delta(t)$$



4. Unit step signal and unit impulse signal

5. Doublet signal

$$f(t)\delta'(t) = f(0)\delta'(t) - f'(0)\delta(t)$$

different from $f(t)\delta(t) = f(0)\delta(t)$

Proof:

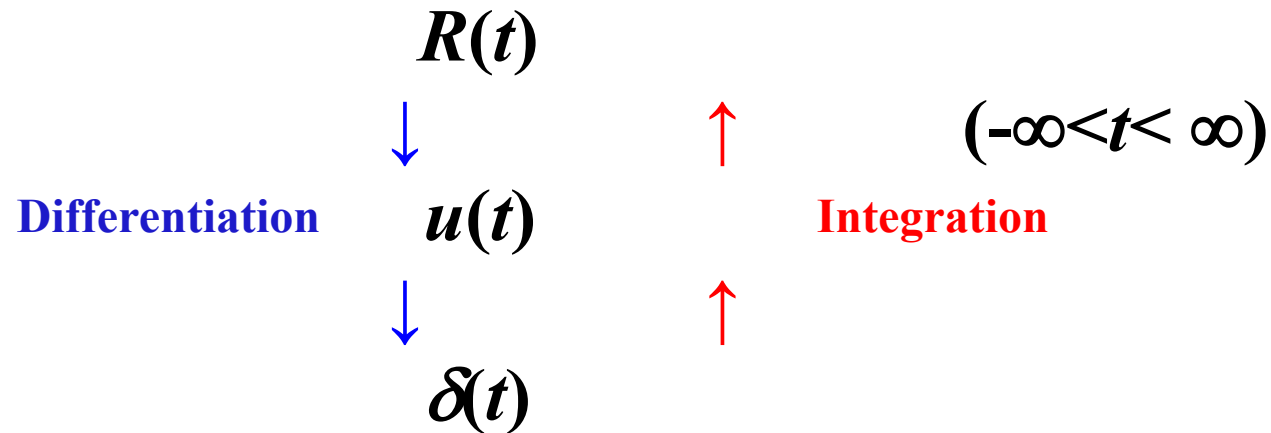
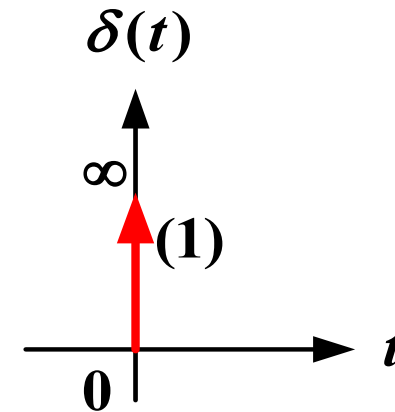
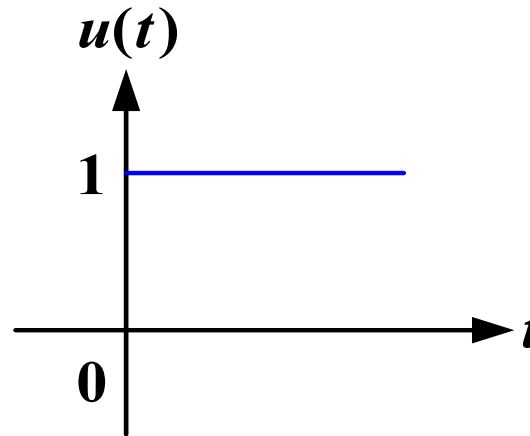
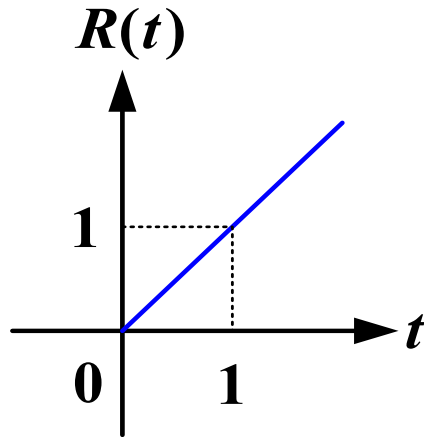
$$[f(t)\delta(t)]' = f(t)\delta'(t) + f'(t)\delta(t)$$

$$\xrightarrow[\delta(t) \text{ --- Sifting}]{}$$

$$[f(0)\delta(t)]' = f(t)\delta'(t) + f'(0)\delta(t)$$

4. Unit step signal and unit impulse signal

Summary



4. Unit step signal and unit impulse signal

Summary

The properties of the unit impulse signal:

(1) Sifting

$$f(t)\delta(t-t_0) = f(t_0)\delta(t-t_0)$$

$$\int_{-\infty}^{\infty} f(t)\delta(t-t_0)dt = f(t_0)$$

$$\int_{-\infty}^{\infty} \delta(t-t_0)dt = 1$$

(2) Parity

$$\delta(-t) = \delta(t)$$

(3) Scaling

$$\delta(at) = \frac{1}{|a|}\delta(t)$$

(4) Calculus Properties

$$\delta(t) = \frac{du(t)}{dt}$$

4. Unit step signal and unit impulse signal

Summary

(5) Doublet

$$\delta'(t) = \frac{d\delta(t)}{dt}$$

$$\delta'(-t) = -\delta'(t)$$

$$\int_{-\infty}^{\infty} \delta'(t) dt = 0$$

$$\int_{-\infty}^t \delta'(\tau) d\tau = \delta(t)$$

$$\int_{-\infty}^{\infty} f(t) \delta'(t) dt = -f'(0)$$

$$f(t) \delta'(t) = f(0) \delta'(t) - f'(0) \delta(t)$$

(6) Convolution Properties

$$f(t) * \delta(t) = f(t)$$

Main Content

- DC component and AC component
- Odd component and even component
- Real part component and imaginary part component

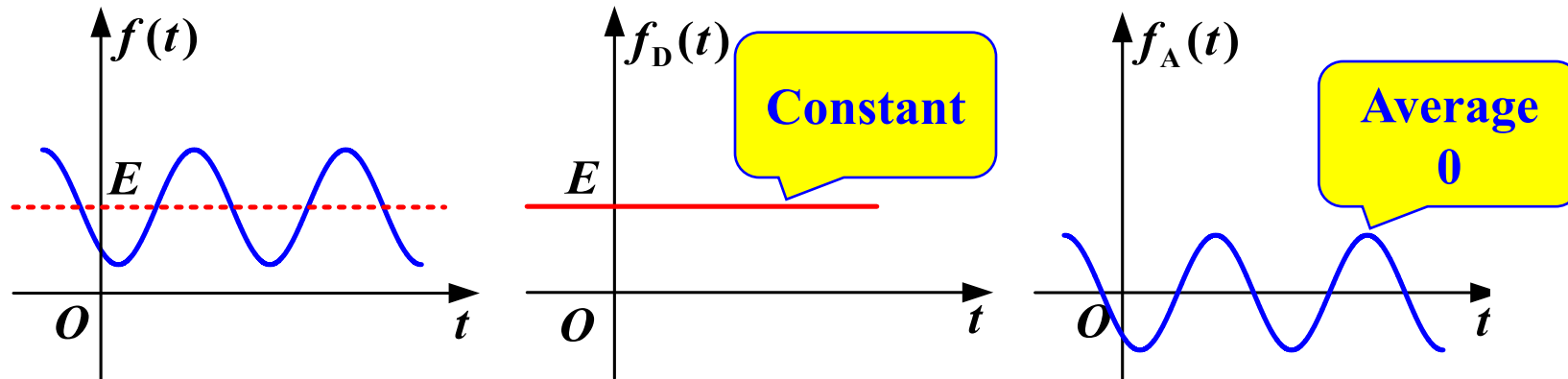
5. Decomposition of signals

1. DC component and AC component

$$f(t) = f_D(t) + f_A(t)$$

$f_D(t)$: DC (direct current component)

$f_A(t)$: AC (alternating current component)



$$\frac{1}{T} \int_{t_0}^{t_0+T} f(t) dt = \frac{1}{T} \int_{t_0}^{t_0+T} [f_A(t) + f_D(t)] dt = f_D(t)$$

DC component, is the average of a signal

5. Decomposition of signals

1. DC component and AC component

Power $P = \frac{1}{T} \int_{t_0}^{t_0+T} f^2(t) dt$

$$= \frac{1}{T} \int_{t_0}^{t_0+T} [f_D(t) + f_A(t)]^2 dt$$

Constant

$$= \frac{1}{T} \int_{t_0}^{t_0+T} f_D^2(t) dt + \frac{2f_D(t)}{T} \cdot \int_{t_0}^{t_0+T} f_A(t) dt$$
$$+ \frac{1}{T} \int_{t_0}^{t_0+T} f_A^2(t) dt$$
$$= f_D^2(t) + \frac{1}{T} \int_{t_0}^{t_0+T} f_A^2(t) dt$$

Signal's Average Power = DC Power + AC Power

5. Decomposition of signals

2. Odd component and Even component

For any real-valued signal:

$$f(t) = f_e(t) + f_o(t)$$

$$\begin{cases} f_e(t) : \text{even component} \\ f_o(t) : \text{odd component} \end{cases}$$

Symmetry:

$$\begin{cases} f_e(t) = f_e(-t) \\ f_o(t) = -f_o(-t) \end{cases}$$

$$f(-t) = f_e(-t) + f_o(-t)$$

$$f(-t) = f_e(t) - f_o(t)$$

Calculation Methods:

$$\begin{cases} f_e(t) = \frac{1}{2}[f(t) + f(-t)] \\ f_o(t) = \frac{1}{2}[f(t) - f(-t)] \end{cases}$$

5. Decomposition of signals

2. Odd component and Even component

Signal Average Power $P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} [f_e(t) + f_o(t)]^2 dt$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f_e^2(t) dt + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f_o^2(t) dt$$
$$+ \lim_{T \rightarrow \infty} \frac{2}{T} \int_{-T/2}^{T/2} f_e(t) f_o(t) dt$$

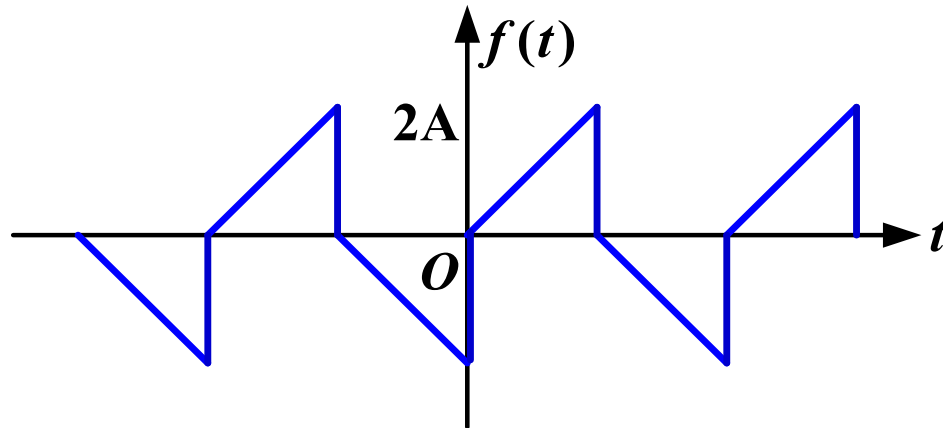
$f_e(t) f_o(t)$ odd function, the integral over a symmetric interval is zero

$$\therefore P = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f_e^2(t) dt + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} f_o^2(t) dt = P_e + P_o$$

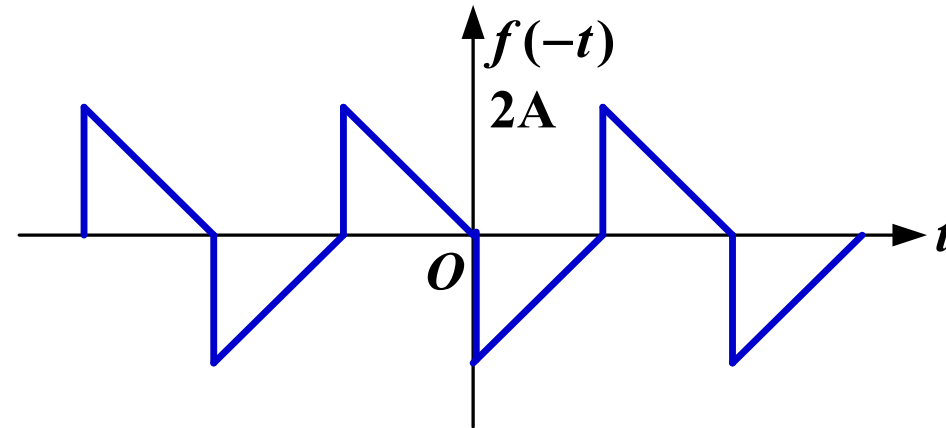
Average power of a signal = Power of **even** component + Power of **odd** component

5. Decomposition of signals

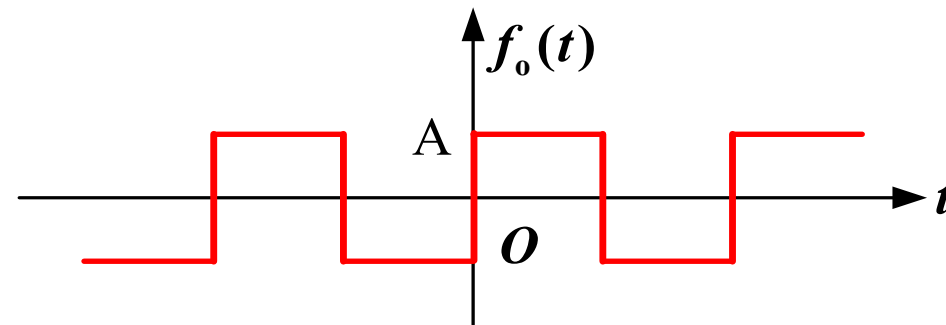
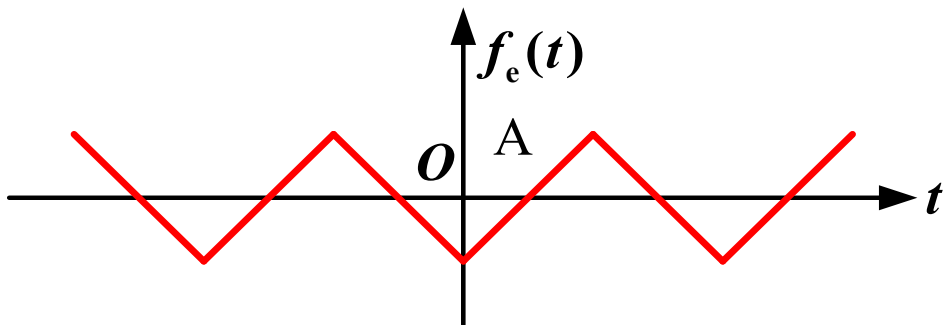
2. Odd component and Even component



$$f_e(t) = \frac{1}{2}[f(t) + f(-t)]$$



$$f_o(t) = \frac{1}{2}[f(t) - f(-t)]$$



3. Real component and imaginary component

A signal with complex instantaneous values can be decomposed into the sum of real and imaginary parts

$$f(t) = f_r(t) + \mathbf{j}f_i(t)$$

Complex conjugate functions

$$f^*(t) = f_r(t) - \mathbf{j}f_i(t)$$

Thus

$$f_r(t) = \frac{1}{2}[f(t) + f^*(t)] \quad \mathbf{j}f_i(t) = \frac{1}{2}[f(t) - f^*(t)]$$

Real-world signals are real-valued, but complex signals can **facilitate the analysis of real signals**

Summary

- **DC component and AC component**
- **Odd component and even component**
- **Real part component and imaginary part component**
- **Chapter 2 Impulse Components**
- **Chapter 3 Sinusoidal Components**